

3. Automate application deployment

Automate application deployment using Jenkins by configuring build triggers, post-build actions, and deployment steps.

1. **Check Docker Installation:** `docker --version`
2. **Check Jenkins Installation:** `jenkins --version`
3. **Check Jenkins Status:** `sudo systemctl status jenkins`
4. **Start Jenkins:** `sudo systemctl start jenkins`
5. **Enable Jenkins at Startup:** `sudo systemctl enable jenkins`
6. **Check Docker Containers:** `docker ps`
7. **Build Docker Image:** `docker build -t my-docker-app .`
8. **Stop Previous Container:** `docker stop my-container || true`
9. **Remove Previous Container:** `docker rm my-container || true`
10. **Deploy New Docker Container:** `docker run -d -p 5000:5000 --name my-container my-docker-app`
11. **Verify Running Container:** `docker ps`
12. **View Container Logs:** `docker logs my-container`
13. **Test Deployed Application:** `curl http://localhost:5000`
14. **Check Git Installation:** `git --version`
15. **Check Git Repository Status:** `git status`
16. **View Git Commit History:** `git log --oneline`

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue



Jenkins



+ New Item

Build History

Build Queue



No builds in the queue.

Build Executor Status

0/2



Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job



Set up a distributed build

Set up an agent



Configure a cloud



Learn more about distributed builds



Ask Gemini

localhost:8081/job/Docker-Deployment/configure

 **Jenkins**

Docker-Deployment

Configure



Configure settings and variables that define the context in which your build runs, like credentials, paths, and global parameters.

☐ Delete workspace before build starts

☐ Use secret text(s) or file(s) ?

☐ Add timestamps to the Console Output

☐ Inspect build log for published build scans

☐ Terminate a build if it's stuck

☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

 Execute shell ?

Command

See [the list of available environment variables](#)

```
cd ~/docker-project
docker build -t my-docker-app .
docker stop my-container || true
docker rm my-container || true
docker run -d -p 5000:5000 --name my-container my-docker-app
```

Advanced ▾

Save

Apply

localhost:8081/job/Docker-Deployment/

 **Jenkins**

/ Docker-Deployment



 Status

 Changes

 Workspace

 Build Now

 Delete Project

 Configure

 Rename

Builds >

No builds

Docker-Deployment

 Add description

Permalinks

REST API

Jenkins 2.568.3

```
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker --version
Docker version 29.7.2, build a7dcaa6
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker rm -f jenkins 2>/dev/null || true
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker volume create jenkins_home
jenkins_home
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker run -d --name jenkins -p 8081:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
Unable to find image 'jenkins/jenkins:lts' locally
lts: Pulling from jenkins/jenkins
d097c8512a96: Pull complete
13045988ecda: Pull complete
91db25d7b900: Pull complete
549414223487: Pull complete
009bab87a9ea: Pull complete
c739d2dd7782: Pull complete
f06c632bcf59: Pull complete
e8d1cd9c12ca: Pull complete
7d8676c4f825: Pull complete
88d27572059a: Pull complete
ba74585817cb: Pull complete
44136fa355b3: Already exists
272717b30638: Download complete
Digest: sha256:c1e4c349365f6d16d88595b2c5f7e8ff39b8ae1d061f62420bac193b4b9616d0
Status: Downloaded newer image for jenkins/jenkins:lts
14a5049451a29759105ee091d7552b50045b9e52d28a92467868fb7a02bd5bfb
ayushi2323@LAPTOP-6LVIDNHT:~/docker-project$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS
14a5049451a2   jenkins/jenkins:lts                "/usr/bin/tini -- /u..." 5 seconds
Up 4 seconds  0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   jenkins
8f3b05f459fb   ae47b29b621b                       "python main.py"         2 hours
Up 2 hours    0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   my-container
7b82b6e69525   nginx                              "/docker-entrypoint..." 2 hours
Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp       my-nginx
```

```
ayushi2323@LAPTOP-6LVIDN X + v
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ ls .github/workflows
docker-ci.yml
ayushi2323 ayushi2323 ayushi2323 2022-4723 15:00 Lcense
ayushi2323 ayushi2323 ayushi2323 2022-4723 15:00 docker-conf.yml
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ cat .github/workflows/do
ocker-ci.yml
on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout source code
        uses: actions/checkout@v4

      - name: Build Docker image
        run: docker build -t my-docker-app .

      - name: Run Docker container
        run: docker run -d -p 5000:5000 --name my-container my-docker-app
        sleep 5

      - name: Test application
        run: curl http://localhost:5000

      - name: Stop container
        run: docker stop my-container
ayushi2323@LAPTOP-6LVIDNNHT:~/docker-project$ git status
On branch main
Your branch is up to date with 'Origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
.github/
```